CHIOMA OKEKE

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EDUCATION

M.S. Computer Science, Arizona State University — Expected May 2027
Mastercard Foundation Scholar | Relevant Coursework: Advanced Operating Systems, Software Security

B.Eng. Computer Engineering, Covenant University — Aug 2022
CGPA: 4.67/5.0 | **Relevant Coursework:** Introduction to Neural Networks, Data Structures & Algorithms

SKILLS

Languages: TypeScript, JavaScript, Python, C++, SQL, Apex.

Frameworks: React.js, Next.js, FastAPI, Node.js, Express.js, Salesforce LWC

Databases: PostgreSQL, MongoDB

Developer Tools: Git, Bitbucket, Linux, Kibana, Jest, Salesforce Code Analyzer

Concepts: Data Structures & Algorithms, OOP, Distributed Systems, CI/CD, REST APIs, RBAC, MCP Integration

EXPERIENCE

EdPlus, Arizona State University - Software Development Assistant — Aug 2025 – Present

- Developing an AI-assisted Salesforce code review workflow combining Salesforce Code Analyzer, custom deterministic checks, and AI analysis across seven quality, security, metadata, and release-readiness gates to improve review consistency and identify risks earlier.
- Expanded an Apex test optimization effort from a single assigned class to approximately **60%** of the existing test suite, raising overall code coverage to **85%** and strengthening release confidence.
- Developed Apex-based backend automation for Salesforce record routing and user reassignment workflows, handling insert and update events to keep records assigned to the appropriate users as availability changed.

Cavista Technologies- Software Engineer I | Healthcare Software — Mar 2023 – Aug 2025

- Engineered a form-to-model workflow in Aurelia and TypeScript that transformed structured form inputs into reusable data models, streamlining the generation of print-ready clinical documentation.
- Shipped a user-facing permissions feature in Aurelia and TypeScript that surfaced mobile-access information for applicable permissions, helping users understand the impact of permission changes before making them.
- Collaborated with backend engineers to diagnose production job failures using Kibana and code-level analysis, reducing investigation time by 35%.

PROJECTS

Laundry Inventory Management System | Web Application

- Built a full-stack inventory and operations platform with RESTful services using Next.js, TypeScript, MongoDB, and Mongoose, centralizing customer, order, item-tracking, and business records across role-based workflows.
- Implemented JWT-based authentication and role-based authorization for Staff/Admin workflows, alongside automated receipt and tag generation, replacing manual recordkeeping and improving staff efficiency by **40%**.

CarbonLint | Carbon-Aware Code Optimizer & Developer Tool

- Built a carbon-aware code optimization tool that identifies inefficient Python code, automatically improves performance, and compares execution time, memory usage, and estimated CO_2 emissions before and after optimization.
- Developed a five-stage analysis and benchmarking engine using Python AST, Radon, and CodeCarbon, implementing four rule-based code transformations and sandboxed execution.
- Integrated a TypeScript VS Code extension and FastMCP server for automated code diagnostics, optimization reports, and AI usage monitoring.

LLM Context Integrity & Prompt Injection Defense | Python / FastAPI / ML / MQTT

- Co-led research investigating prompt-injection and context-integrity attacks against LLM-enabled smart-home systems, evaluating direct, indirect, and tool-integrated attack scenarios.
- Evaluated a three-layer defense architecture combining embedding-based detection, input hardening, and deterministic output validation to study how layered controls reduce unsafe LLM-driven device actions.